DESIGNING AN EBAY-LIKE DATA WAREHOUSE

“Fact Finders' Blueprint for Business Intelligence”

IST 722 Data Warehouse

Team Name: Fact Finders

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1. Project Overview

Introduction

The vBay Data Warehouse project is a comprehensive initiative aimed at unlocking business intelligence for an online auction platform by converting raw, fragmented data into a unified, analytics-ready format. Centered around key auction operations such as bidding, item listings, and transaction outcomes, this project uses dimensional modeling and ETL processes to design a scalable star schema. The resulting data warehouse powers a Power BI dashboard that reveals trends in bidder engagement, item demand, and auction performance across categories and cities.

Key Objectives

- Architect a dimensional data model that reflects core auction business processes: bidding, item listing, and auction closure.
- Execute ETL workflows to extract, cleanse, and transform datasets including bids, users, items, ratings, and regional ZIPs.
- Populate fact tables and conforming dimensions to ensure consistent, historical tracking of auctions.
- Visualize insights in Power BI through KPIs, charts, and filters for stakeholders to explore performance by category, location, and user.
- Enable evidence-based decision-making for improving auction efficiency, pricing strategies, and platform experience.

Business Value

The dashboard built on top of the vBay data warehouse directly supports key business questions, offering:

- **Bidder Engagement Insights:** By showcasing user-wise bid amounts and frequency, the system helps identify top bidders and their behavioral patterns.
- **Auction Success Metrics:** The 17.65% success rate highlights improvement areas in pricing or targeting strategies.
- **Category-Level Performance:** High unsold rates in categories like “Collectables” and “Antiques” prompt inventory reassessment.
- **Regional Trends:** Analysis of cities like North P. and Chattanooga helps tailor regional campaigns and support initiatives.

These insights promote operational efficiency by surfacing both high-performing patterns and underperforming segments.

Strategic Value

At a strategic level, the vBay data warehouse enables the platform to shift from reactive reporting to proactive strategy:

- Optimizing Listings: Data on unsold categories and item types can help sellers tailor offerings and reduce auction failures.
- Customer Intelligence: User-level bid tracking informs loyalty programs and segmentation strategies.

- Regional Targeting: Location-wise bid and rating analysis supports geo-personalized campaigns and demand forecasting.
- Scalability & Advanced Analytics: The architecture lays the groundwork for predictive modeling—such as forecasting successful auctions or identifying high-value users early in the bidding cycle.

By embedding intelligence into daily auction decisions, vBay gains a lasting competitive advantage in the online marketplace.

2. Team Structure

Team Members and Roles

- **Saarthak Joshi – Data Warehouse Architect & ETL Developer**
  Saarthak led the end-to-end warehouse design, from dimensional modeling to implementation. He developed the star schema that accurately reflects the vBay auction system's business processes. Additionally, he handled the extraction, transformation, and loading (ETL) of raw data into fact and dimension tables using SQL.
- **Tejas Mistry – Data Engineer & BI Analyst**
  Tejas focused on data profiling, transformation logic, and validation. He also developed the Power BI dashboard, designing intuitive visualizations and KPIs that offer insights into user behavior, auction performance, and bidding trends.

Collaboration Approach

Despite being a two-member team, Saarthak and Tejas efficiently divided responsibilities based on technical strengths. Regular checkpoints were held to review schema integrity, validate ETL outcomes, and refine BI visualizations. The team emphasized iterative testing and continuous improvement to maintain data accuracy and alignment with business goals.

Project Charter

The goal of this project was to design a scalable, business-driven data warehouse and develop actionable business intelligence dashboards for an eBay-like online auction platform. The team was committed to building a robust backend architecture that could power real-time insights and historical analysis for vBay stakeholders.

3. Project Plan

Development Phases and Timeline

The vBay Data Warehouse project was executed in three structured phases, each focusing on a core component of the data pipeline and analytical lifecycle:

- **Phase 1: Requirement Analysis & Dimensional Modeling**
    - Identified key business processes: Bidding, Item Listing, and Auction Completion
    - Developed a Bus Matrix to map fact and dimension tables
    - Defined table granularity and identified key metrics (e.g., bid amounts, success rate)
- **Phase 2: Data Integration & Warehouse Implementation**
    - Performed ETL: Extracted, cleaned, and transformed datasets including bids, items, users, ratings, and zip codes
    - Constructed fact tables: fact_bids, fact_items, and fact_auction_status
    - Populated dimension tables: dim_bidder, dim_item, dim_time
    - Validated data consistency, resolved duplicates, handled nulls, and normalized formats
- **Phase 3: Business Intelligence and Insights Delivery**
    - Designed an interactive Power BI dashboard
    - Created KPIs such as average bid amount, success rate %, and unsold item trends
    - Added visual filters for item type and city to enhance user interactivity
    - Interpreted insights to align with business questions and stakeholder needs

The vBay Data Warehouse project was executed through a structured, three-milestone framework aligned with IST 722 course requirements. Each milestone delivered a core component of the data warehouse lifecycle—from conceptual modeling to business intelligence.

🧩 **Milestone 01 – Bus Matrix Design**

- Identified key business processes: Bidding, Item Listing, and Auction Completion
- Defined fact tables (`fact_bids`, `fact_items`, `fact_auction_status`) and their associated dimensions (`dim_bidder`, `dim_item`, `dim_time`)
- Created a Bus Matrix outlining process-to-dimension relationships and table granularity
- Delivered: Slide deck with Bus Matrix, process overview, and schema rationale

🛠️ **Milestone 02 – Dimensional Model & Implementation (1 Process Minimum)**

- Developed the star schema for the Bidding process using `fact_bids`
- Loaded and cleaned data for related dimensions: `dim_bidder` and `dim_time`
- Executed ETL pipelines using SQL to transform raw vBay datasets into structured warehouse tables

- Validated integrity through row count checks, null handling, and key consistency
- Delivered: Presentation covering detailed schema, data transformations, and implemented SQL scripts

📊 **Milestone 03 – BI Demonstration**

- Designed and deployed an interactive Power BI dashboard
- Visualized auction success rates, user bidding trends, category performance, and city-level analysis
- Integrated filters (e.g., item type, city) and KPIs (e.g., Avg Bid Amount, Success Rate %, Total Bids)
- Provided business value discussion with interpretation of trends
- Delivered: Slide deck with demo visuals, KPIs, and real-world business implications

Project Management Approach

The team followed a milestone-driven development cycle with clear responsibilities and deadlines. Regular collaboration checkpoints ensured synchronization between modeling, ETL, and visualization efforts. Documentation, testing, and version control were practiced throughout to ensure smooth transitions between milestones.

4. Data Overview

Source Data Description

The vBay dataset represents an auction-based e-commerce platform. It captures end-to-end interactions between bidders and item listings, allowing analytical modeling of auction success, user behavior, and category-level trends. The primary source tables used in this project include:

- **Bids:** Transactional bid records, including amount, bid time, and user association.
- **Items:** Metadata on auction listings such as starting price, final sale price, category, and duration.
- **Auction Status:** Periodic snapshots of listing status, inventory, and reorder activity.
- **Users:** Bidder details, location, and ZIP code-level demographics.
- **Ratings:** Buyer and seller ratings that reflect engagement quality and trust signals.
- **Zipcodes:** Mapped regional data for user segmentation and geographic insights.

Data Profiling Insights

Prior to loading into the dimensional model, the data was extensively profiled and cleaned:

- Null ZIP codes and blank rating values were identified and addressed.
- Duplicate user and bid records were removed using composite keys.
- Inconsistent bid timestamps and auction durations were normalized.
- Region and city mappings were enriched using ZIP-to-city lookups.

Bids

Attribute	Details
Table Type	Business Process (Transactional)
Row Count	236
Business Key	Composite (Bid ID or Bidder ID + Item ID + Timestamp)
One Row Represents	A unique bid placed by a user
Key Columns	Bid Amount, Bid Time, Bidder ID, Item ID
Observations	Valid timestamps, varying bid values by category and region

Items

Attribute	Details
Table Type	Accumulating Snapshot
Row Count	92
Business Key	Item ID
One Row Represents	A single item listed for auction
Key Columns	Start Price, Final Price, Item Type, Listing Duration
Observations	Some items received no bids (used for unsold analysis)

Auction Status

Attribute	Details
Table Type	Periodic Snapshot
Row Count	Matches items table
Business Key	Item ID + Time Period
One Row Represents	Status of an item at a particular interval
Key Columns	Inventory Status, Reorder Count
Observations	Tracks inventory movement and supply side analytics

Users

Attribute	Details
Table Type	Master Key
Row Count	50
Business Key	User ID
One Row Represents	A unique user/bidder

Attribute	Details
Key Columns	User ID, ZIP Code
Observations	Some ZIPs missing or malformed; enriched via ZIP mapping

Ratings

Attribute	Details
Table Type	Master Key
Row Count	64
Business Key	User ID
One Row Represents	A user's aggregated rating
Key Columns	Rating Value, User ID
Observations	Used to evaluate bidder quality and trust level

Zipcodes

Attribute	Details
Table Type	Reference/Lookup
Row Count	10+ unique ZIP-city pairs
Business Key	ZIP Code
One Row Represents	A unique city/region
Key Columns	ZIP, City, Region
Observations	Mapped to enable city-wise filtering in the BI dashboard

Challenges and Assumptions

- Assumed one-to-one mapping between ZIP and City for region segmentation.
- Some auctions lacked final sale price; treated as unsold during transformation.
- Usernames were assumed unique where User IDs were not available.
- Rating granularity varied and was normalized for comparative analysis.

5. Dimensional Modeling

High-Level Design (Bus Matrix)

Business Process Name	Fact Table	Fact Grain Type	Granularity	Facts	dim_bidder	dim_item	dim_status	dim_time
					Dimension Tables			
Bidding	fact_bids	Transaction	One row per bid placed	bid_amount, bid_time	x	x	x	x
Item Listing	fact_items	Accumulating Snapshot	One row per listed item	listing_price, final_price, num_bids		x		x
Auctiom Complete	fact_auction_status	Periodic Snapshot	winning_bid, num_bidders, auction_duration	inventorylevel, reordercount	x	x	x	x

The business processes identified include Bidding, Item Listings, and Auction Completion.

FACT TABLE AND DIMENSION TABLE

Detail each fact table along with its granularity and the metrics it captures:
- `fact_bids`: Transactions, captures bid amount, bid time.
- `fact_items`: Accumulating Snapshot, captures listing price, final price, number of bids.
- `fact_auction_status`: Periodic Snapshot, captures inventory level, reorder count.

Dimension Tables

Dimension	Description	Key Attributes
dim_bidder	Contains bidder information	Bidder ID, ZIP Code, City
dim_item	Item catalog with categories and types	Item ID, Item Type, Category, Listing Duration
dim_time	Supports time-based analysis for all facts	Date ID, Day, Month, Quarter, Year

Design Highlights and Standards

- **Schema Style**: Classic **star schema** for simplicity, performance, and intuitive reporting.
- **Surrogate Keys**: Used for all dimension tables to maintain warehouse integrity.
- **Naming Conventions**:
  - Dimensions prefixed with dim_ (e.g., dim_item)
  - Facts named by process (e.g., fact_bids)
  - camelCase used for column naming (e.g., bidAmount, finalPrice)

6. Data Warehouse Implementation

Warehouse Platform

The vBay data warehouse was implemented using **Snowflake**. The architecture follows a traditional three-tier structure: **Raw Layer**, **Staging Layer**, and **Star Schema Layer**.

ETL Pipeline Overview

The ETL process was executed using a combination of **SQL scripts** and **manual preprocessing** (e.g., Excel and Python for ZIP mapping). Key steps in the pipeline included:

- **Extract**:
  Data was sourced from CSV files representing tables such as vb_bids, vb_items, Users, Ratings, and Zipcodes.
- **Transform**:
  - Cleaned missing values (e.g., ZIP codes, blank bids)
  - Normalized timestamps for bid and listing durations
  - Calculated derived fields like bid count and success rate
  - Mapped ZIP codes to city and region using lookup enrichment
  - Aggregated unsold items for category-level analytics
- **Load**:
  - Populated dimension tables: dim_bidder, dim_item, dim_time
  - Loaded fact tables: fact_bids, fact_items, fact_auction_status
  - Maintained referential integrity via surrogate keys and lookups

CODE:

1. Setup: Data Warehouse, Database, and Raw Schema

```
-- Create the warehouse
CREATE WAREHOUSE my_warehouse;

-- Create the database
CREATE DATABASE auctions_db;

-- Create the schema
CREATE SCHEMA auction_db.raw;
```

🧱 2. Table Creation – Raw Layer (Staging Zone)

```
-- Users table
CREATE TABLE auction_db.raw.Users (
    user_id INT,
    user_email STRING,
    user_firstname STRING,
    user_lastname STRING,
    user_zip_code INT
);
```

```sql
-- Bids table
CREATE TABLE auction_db.raw.Bids (
    bid_id INT,
    bid_user_id INT,
    bid_item_id INT,
    bid_datetime STRING,
    bid_amount FLOAT,
    bid_status STRING
);

-- Items table
CREATE OR REPLACE TABLE auction_db.raw.items (
    item_id INT,
    item_name STRING,
    item_type STRING,
    item_description STRING,
    item_reserve FLOAT,
    item_enddate STRING,
    item_sold INT,
    item_seller_user_id INT,
    item_buyer_user_id FLOAT,
    item_soldamount FLOAT
);

-- Ratings table
CREATE TABLE auction_db.raw.Ratings (
    rating_id INT,
    rating_by_user_id INT,
    rating_for_user_id INT,
    rating_astype STRING,
    rating_value INT,
    rating_comment STRING
);

-- Zipcodes table
CREATE TABLE auction_db.raw.Zipcodes (
    zip_code INT,
    zip_city STRING,
    zip_state STRING,
    zip_lat FLOAT,
    zip_lng FLOAT
);
```

📈 3. Business Problem: Auction Success Rate

```
-- Calculates % of successful auctions (item sold and sold amount
≥ reserve)
SELECT
    COUNT(*) AS total_auctions,
    SUM(CASE WHEN item_sold = 1 AND item_soldamount >=
item_reserve THEN 1 ELSE 0 END) AS successful_auctions,
    ROUND(100.0 * SUM(CASE WHEN item_sold = 1 AND item_soldamount
>= item_reserve THEN 1 ELSE 0 END) / COUNT(*), 2) AS
success_rate_percentage
FROM auction_db.raw.items;
```

🧑‍💻 4. Business Problem: User Behavior and Bid Engagement

```
-- Top 10 bidders by total bids and their average bid amount
SELECT
    u.user_id,
    u.user_firstname || ' ' || u.user_lastname AS bidder_name,
    COUNT(b.bid_id) AS total_bids,
    ROUND(AVG(b.bid_amount), 2) AS avg_bid_amount
FROM auction_db.raw.Bids b
JOIN auction_db.raw.Users u ON b.bid_user_id = u.user_id
GROUP BY u.user_id, u.user_firstname, u.user_lastname
ORDER BY total_bids DESC
OFFSET 0 ROWS FETCH NEXT 10 ROWS ONLY;
```

🗂️ 5. Business Problem: Item Type Popularity

```
-- Number of bids per item type
SELECT
    i.item_type,
    COUNT(b.bid_id) AS total_bids
FROM auction_db.raw.Bids b
JOIN auction_db.raw.items i ON b.bid_item_id = i.item_id
GROUP BY i.item_type
ORDER BY total_bids DESC;
```

🏙️ 6. Business Problem: City-Wise Bidding & Rating Activity

```
-- Total bids by city and associated user ratings
SELECT
    z.zip_city,
    r.rating_value,
    COUNT(b.bid_id) AS total_bids
FROM auction_db.raw.Bids b
JOIN auction_db.raw.Users u ON b.bid_user_id = u.user_id
JOIN auction_db.raw.Zipcodes z ON u.user_zip_code = z.zip_code
JOIN auction_db.raw.Ratings r ON u.user_id = r.rating_for_user_id
GROUP BY z.zip_city, r.rating_value
ORDER BY total_bids DESC;
```

📦 7. Business Problem: Sold vs. Unsold Items by Category

```
-- Auction outcome stats by item type
SELECT
    item_type,
    ROUND(AVG(item_reserve), 2) AS avg_reserve_price,
    ROUND(AVG(item_soldamount), 2) AS avg_sold_price,
    COUNT(CASE WHEN item_sold = 1 THEN 1 END) AS sold_count,
    COUNT(CASE WHEN item_sold = 0 THEN 1 END) AS unsold_count
FROM auction_db.raw.items
GROUP BY item_type
ORDER BY sold_count DESC;
```

🕒 8. Business Problem: Bidding Activity Over Time

```
-- Tracks total bids per item and shows when the auction ends
SELECT
    i.item_id,
    i.item_type,
    COUNT(b.bid_id) AS total_bids,
    i.item_enddate
FROM auction_db.raw.items i
JOIN auction_db.raw.Bids b ON i.item_id = b.bid_item_id
GROUP BY i.item_id, i.item_type, i.item_enddate
ORDER BY i.item_enddate;
```

7. Business Intelligence Implementation

Tool Used

Power BI was used to create a visually interactive dashboard that delivers key insights on user behavior, bidding activity, item trends, and auction performance.

Dashboard Highlights

The dashboard focuses on surfacing metrics that are critical to understanding platform dynamics:

- **Average Bid Amount**: ₹1.73K — representing general bidder activity levels across the platform.
- **Success Rate Percentage**: 17.65% — tracks the ratio of completed auctions to total listings.
- **Total Bids and Ratings**: 236 bids and 64 ratings were analyzed for behavior and credibility evaluation.

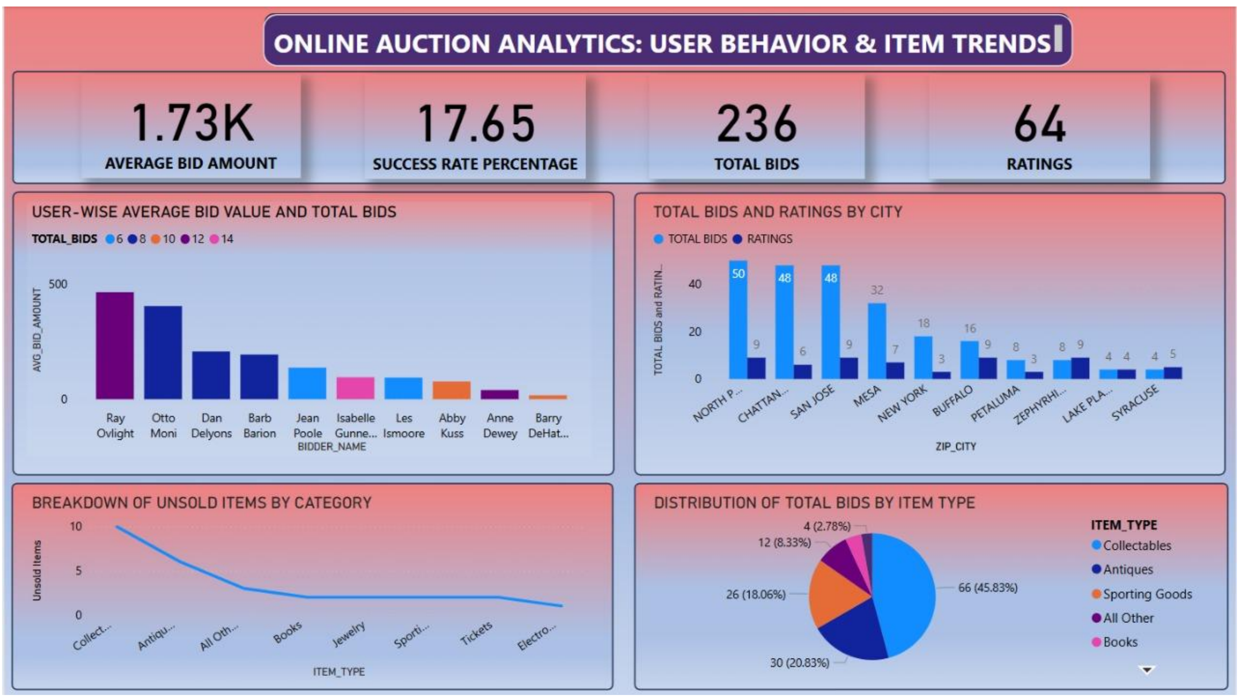
Key Visual Insights

- **User-wise Bidding Trends**: Bar chart shows average bid values and total bids per user, highlighting top bidders like Ray Ovlight and Otto Moni.

- **Bids and Ratings by City**: Cities such as North P., Chattanooga, and San Jose have high bid volume and engagement, supporting regional segmentation strategies.
- **Unsold Items by Category**: A line chart highlights item types (e.g., Collectables, Antiques) with the highest unsold rates—critical for inventory optimization.
- **Bids by Item Type**: Pie chart reflects bid distribution across categories; Collectables dominate with 45.83% of total bids.

Interactivity Features

Filters by item type and city allow dynamic exploration of auction data, helping stakeholders drill down into performance by region or product category.



8. Challenges and Solutions

1. Incomplete or Inconsistent ZIP Code Data

- Challenge: Several user records had missing or malformed ZIP codes, which affected regional and city-level analysis.
- Solution: Implemented a ZIP code enrichment process by mapping valid ZIPs to city names using a reference table. Records without any valid ZIP were flagged and excluded from location-based dashboards.

2. Multiple Bids Per User Without Clear Unique Keys

- Challenge: The bids dataset lacked a consistent primary key. Multiple bids from the same user on the same item made it difficult to uniquely identify rows.
- Solution: Constructed a surrogate bid key using a combination of user ID, item ID, and bid timestamp. Ensured uniqueness by validating composite key combinations.

3. Varying Auction Outcomes (No Final Price / Unsold Items)

- Challenge: Several items had no final price, leading to ambiguity in determining auction success or failure.
- Solution: Defined a clear business rule: if no final price or bid exists, the item is classified as unsold. This rule was consistently applied during transformation and used in success rate calculations.

4. Sparse or Imbalanced Ratings Data

- Challenge: The ratings dataset had fewer entries compared to users or bids, with some users unrated or inconsistently rated.
- Solution: Normalized the rating scale, removed outliers, and treated missing ratings as "Not Rated" in BI visualizations. Added a filter in Power BI to exclude unrated users when analyzing engagement quality.

5. Fact Table Integration and Temporal Alignment

- Challenge: Linking bids, items, and auction statuses by time required careful temporal alignment, especially with snapshot-based fact tables.
- Solution: Constructed a dim_time table to standardize date representations and ensured all timestamps were converted to a uniform format during ETL.

6. Dashboard Performance and Visual Clarity

- Challenge: The Power BI dashboard initially displayed inconsistent trends due to raw data anomalies and excessive visual clutter.
- Solution: Refined the visual layout with clear KPIs and filters. Applied DAX logic to clean up null results, and reduced visual overload by grouping metrics logically (e.g., success rate, bid trends, city breakdown).

9. Reflection and Next Steps

Lessons Learned

- **Early Role Alignment Matters**: With a compact team, clearly defining responsibilities early in the project helped streamline progress, avoid duplication of efforts, and ensure accountability throughout each milestone.
- **ETL Testing Is Crucial**: Iterative testing of ETL scripts at every stage—especially after transformation logic—prevented downstream data quality issues and reinforced the value of early validation.
- **Data Profiling Enables Better Modeling**: Thorough initial data profiling revealed patterns (e.g., unsold items, ZIP code gaps) that shaped how we modeled dimensions and chose relevant KPIs.

Future Enhancements

- **Predictive Analytics**: Integrating machine learning models to forecast auction success likelihood, expected final prices, or bidder activity trends could elevate the strategic value of the warehouse.
- **Supply Chain Metrics**: Expanding the schema to include delivery time, item shipment tracking, and seller fulfillment metrics would support more holistic e-commerce analysis.
- **Bidder Segmentation**: Introducing bidder segmentation using behavioral clustering could support targeted engagement or personalized auction recommendations.
- **Real-Time BI Capabilities**: Implementing streaming data ingestion for live bidding events and auction updates would allow real-time dashboarding and alerts.